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Molecular dynamics and machine learning framework for predicting ion transport and mechanical properties of ionic liquid@polyvinylidene fluoride gel polymer electrolyte

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ABSTRACT

Advancing high-performance gel polymer electrolytes (GPEs) is critical for the development of safe, efficient, and mechanically robust lithium-ion batteries (LIBs). In this work, we present a synergistic framework that integrates molecular dynamics (MD) simulations with machine learning (ML) to predict and rationalize the ionic conductivity and mechanical properties of ionic liquid@polyvinylidene fluoride (IL@PVDF)-based GPEs. A comprehensive analysis of MD-derived descriptors, including the lithium-ion diffusion coefficient, density, Pugh's ratio, and elastic moduli, reveals the underlying structure–property relationships. Benchmarking five ML regressors using log-transformed conductivity data identifies Extreme Gradient Boosting (XGB) as the most accurate, generalizable, and robust predictor. SHapley Additive exPlanations (SHAP) analysis provides mechanistic interpretability, highlighting the dominant roles of ion diffusion and matrix flexibility in governing conductivity. Furthermore, SHAP interaction plots uncover nonlinear synergies between ion mobility and ductility, demonstrating that transport and mechanical performance can be simultaneously optimized. Collectively, this MD–ML framework offers a predictive, interpretable, and scalable strategy to accelerate the rational design of next-generation IL-based GPEs, thus providing actionable guidance for high-throughput screening and advanced electrolyte engineering in LIBs.

Introduction

Lithium-ion batteries (LIBs) have become the dominant energy storage technology for a wide range of applications, including portable electronics, electric vehicles, aerospace systems, and renewable energy storage [1–3]. Their high energy density, high capacity, and good charge–discharge efficiency make them ideal for both consumer and industrial use. However, safety, and longevity of LIBs are significantly influenced by the properties of the electrolyte, which governs lithium-ion transport, electrochemical stability, and interfacial compatibility with electrodes [4,5]. Recent developments in solid-state battery architectures have further enhanced performance of LIBs, offering improved ionic conductivity, thermal stability and cycle life [6–9]. As the demand for sustainable and compact energy solutions grows, LIBs continue to play a critical role in enabling next-generation energy systems.

The electrolyte is a critical component in LIBs, directly influencing key performance metrics such as ionic conductivity, electrochemical stability, and operating voltage range. While traditional liquid electrolytes are widely used due to their high ionic mobility, they suffer from several drawbacks, including flammability, leakage, and limited electrochemical stability windows that raise significant safety concerns, particularly for emerging applications in flexible, wearable, and high-temperature devices [10–12]. To overcome these limitations, solid-state electrolytes, and in particular, gel polymer electrolytes (GPEs), have attracted considerable attention [13–15]. GPEs offer numerous advantages, including enhanced thermal and electrochemical stability, reduced volatility, and improved mechanical processability [16,17]. They consist of a solid polymer matrix that physically entraps or solvates liquid electrolytes, maintaining ionic mobility while suppressing leakage and dendrite formation [18]. This dual functionality enables GPEs to serve as the electrolyte, separator, and binder within a battery cell. Commonly used polymer hosts include poly(methyl methacrylate) (PMMA), poly(vinylidene fluoride) (PVDF), poly(vinylidene fluoride-co-hexafluoropropylene) (PVDF-HFP), polyacrylonitrile (PAN), and polyethylene oxide (PEO), which offering distinct advantages in terms of mechanical strength, compatibility with lithium salts, and thermal stability [19–21].

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Extensive studies have shown that LiF-rich interphases are crucial for stabilizing electrode interfaces, owing to their high Young's modulus, elevated surface energy, and favorable Li-ion transport characteristics [22–24]. Traditional strategies to enrich the solid-electrolyte interphase (SEI) with LiF have largely focused on increasing the fluorine content in electrolytes. Although high-concentration electrolytes (HCEs) and localized high-concentration electrolytes (LHCEs) can generate LiF-rich SEI layers in situ, their high cost limits practical application [25]. In this context, the incorporation of fluorine-containing additives (F-additives) at low concentrations (<10 vol%) into conventional carbonate electrolytes offers a more feasible route. The effectiveness of these additives largely depends on their F/C molar ratio, with higher ratios promoting more robust LiF formation at the lithium metal interface [26].

Among various polymer hosts, polyvinylidene fluoride (PVDF)-based systems combined with ionic liquids (ILs) have emerged as highly promising candidates for the development of next-generation GPEs in LIBs [27–31]. PVDF is a multifunctional polymer known for its outstanding thermal and electrochemical stability, strong mechanical properties, high dielectric constant, excellent chemical resistance, and flexibility [32,33]. These characteristics make it an excellent choice as a matrix material for safe, flexible, and long-lasting energy storage devices, offering high anodic stability and good compatibility with liquid electrolytes. Ionic liquids (ILs), which are room-temperature molten salts, serve as versatile and tunable solvents or plasticizers within the GPEs [28]. Their wide electrochemical stability windows, high intrinsic ionic conductivity, low vapor pressure, and non-flammable nature make them especially attractive for enhancing the safety and electrochemical performance of polymer electrolytes [34,35]. Thus, the synergy between PVDF and ILs enables the formation of GPEs that exhibit improved ion mobility, interfacial compatibility, and long-term operational stability, making them excellent candidates for high-performance, safe, and flexible lithium-ion battery applications [36].

Recent research in GPEs has increasingly focused on enhancing mechanical robustness and electrochemical efficiency to meet the demands of next-generation energy devices. For instance, Jie et al. [37] reported a PVDF-HFP-based gel polymer electrolyte using N-methyl-2-pyrrolidone (NMP) as a safe solvent, achieving high ionic conductivity (7.24×10^{-4} S/cm) and stable cycling over 500 h, demonstrating its suitability for safe LIBs. Jelani et al. [38] reported an imidazolium-based ionic liquid-infused polyvinyl alcohol (PVA) polymer electrolyte exhibiting an impressive ionic conductivity of 1.52×10^{-3} S/cm at room temperature, highlighting its potential for high-performance energy storage and light-emitting electrochemical devices. Furthermore, More et al. [39] reported a solvate ionic liquid-based gel polymer electrolyte supported by an imidazolium-functionalized copolymer, achieving a notable ionic conductivity of 1.0×10^{-4} S/cm and a compressive elastic modulus of 0.100 MPa at room temperature. This system also exhibited a lithium-ion transference number of 0.71, demonstrating excellent electrochemical performance and dendrite suppression, making it a strong candidate for stretchable LIBs. Building upon these advances, Li et al. [40] introduced a triadic molecular synergy strategy in ionic liquid-based flame-retardant quasi-solid polymer electrolytes, decoupling lithium-ion transport mechanisms while simultaneously enhancing mechanical stability and electrochemical performance, achieving a high lithium-ion transference number (0.72) and improved cycling stability in Li||$LiFePO_4$ and Li||$LiNi_{0.8}Co_{0.1}Mn_{0.1}O_2$ cells, further demonstrating the potential of molecular engineering approaches for next-generation electrolytes.

Molecular dynamics (MD) simulations offer atomic-level insights into ion transport, polymer segmental dynamics, and solvation environments under realistic thermodynamic conditions, making them an invaluable tool for evaluating key electrolyte properties such as diffusion coefficients, density, and mechanical moduli [41–43]. In the context of GPEs for LIBs, MD has played a crucial role in elucidating Li^+ transport mechanisms, polymer–solvent interactions, and the impact of ionic liquid incorporation on both ionic conductivity and structural flexibility [44]. For example, Shen et al. [45] showed through molecular dynamics simulations that plasticizing a novel single-ion conducting gel polymer electrolyte significantly enhanced lithium-ion transport, increasing the diffusion coefficient from 8.05×10^{-11} cm/s to 1.37×10^{-9} cm2/s, thereby facilitating improved ionization and mobility within the polymer matrix. In another work, Du et al. [46] demonstrated via molecular dynamics simulations that lithium-ion transport within their non-flammable gel polymer electrolyte is primarily governed by intra-chain hopping mechanisms facilitated by the three-dimensional (3D) interpenetrating polymer network. This structure enhances ionic mobility by providing continuous conduction pathways, contributing to the system's high electrochemical performance and improved safety.

In recent years, machine learning (ML) has emerged as a transformative tool in materials science, offering powerful capabilities to accelerate the design and optimization of electrolytes for advanced energy storage systems, including LIBs [47–50]. Specifically, its application to the rational design of GPEs remains relatively nascent yet rapidly expanding. By leveraging comprehensive datasets and advanced predictive algorithms, ML enables the rapid identification of optimal polymer matrices, ionic liquid plasticizers, and processing parameters, thereby reducing reliance on traditional trial-and-error approaches. These data-driven methodologies substantially decrease the need for time-consuming and costly experimental procedures while enabling exploration of a broader materials design space. This capability is particularly valuable for developing next-generation electrolyte systems that are mechanically flexible, thermally stable, and electrochemically efficient, tailored for solid-state lithium-ion batteries and other advanced energy-storage technologies. Moreover, by training ML models on MD-derived datasets, researchers can rapidly and accurately predict key performance metrics such as ionic transport and mechanical stability in novel electrolyte formulations, thereby accelerating materials screening and optimization [51].

In this work, we present a comprehensive study on the modulation of ion transport and mechanical properties of ionic liquid-based gel polymer electrolytes (IL@PVDF-based GPEs), with a specific focus on their application in LIBs. To inform our machine learning workflow and guide descriptor selection, we utilized a curated dataset of 103 IL@PVDF complexes, systematically constructed using a fixed alaninate anion and PVDF polymer matrix while varying the cationic components to ensure chemical diversity and structural complexity. The computational strategy is summarized in Fig. 1. MD simulations were employed to extract key parameters, including diffusion coefficients, ionic conductivity, density, Young's modulus, shear modulus, bulk modulus, and Pugh's ratio. These simulation-derived descriptors were then used to train robust ML models for accurate and rapid property prediction across an expanded materials space. This integrated MD–ML framework not only provides atomistic insights into IL–polymer interactions and dynamic behavior but also establishes a predictive platform for designing thermally stable, mechanically resilient, and highly conductive GPEs. Ultimately, this approach accelerates the discovery and rational design of next-generation electrolytes tailored for flexible, safe, and high-performance LIBs.

Computational methods

MD simulations

MD simulations were performed using the Forcite module in BIOVIA Materials Studio [52] to explore the structural, ion transport, and mechanical properties of IL@PVDF complexes relevant to LIB applications. Each simulation system was initially constructed using the Amorphous Cell module by embedding a fixed alaninate anion within a PVDF matrix (with a repeating unit of eight), while systematically varying the cationic components across 103 distinct ionic liquids, covering 10 different cation families to ensure wide chemical diversity. The initial amorphous structures were generated using Amorphous Cell

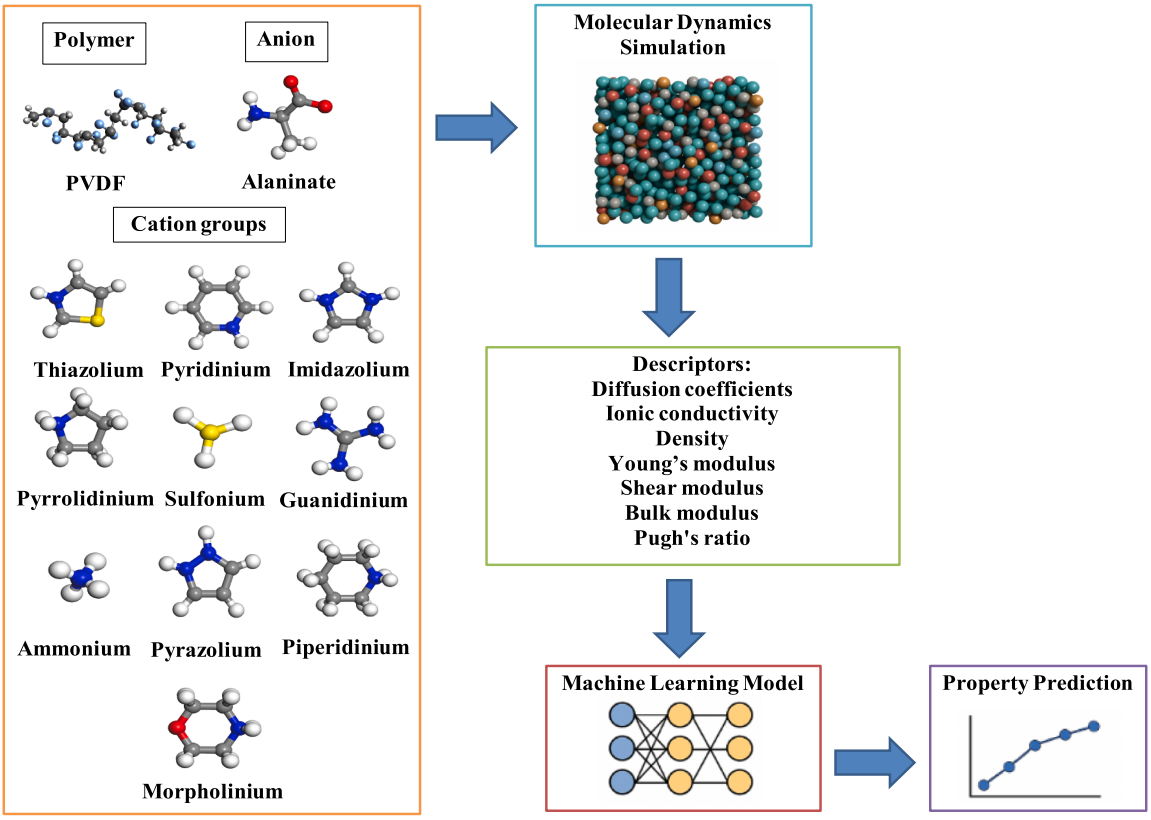


Fig. 1. Schematic overview of the integrated MD and ML workflow developed to predict ion transport and mechanical properties of IL@PVDF-based GPEs for LIBs.

calculations with cubic cell dimensions of 10.60 × 10.60 × 10.60 Å^3 and cell angles α = β = γ = 90°. These parameters were selected to provide physically reasonable densities and to prevent artificial molecular overlaps during the packing stage. Five representative IL@PVDF configurations, each containing a different cation, are presented in Fig. S1 to illustrate the molecular diversity and structural setup of the MD simulations. After construction, all systems underwent energy minimization using the Ensemble Configuration Annealing method and the COMPASS force field [53]. The systems were equilibrated under both canonical (NVT) and isothermal–isobaric (NPT) ensembles. Temperature control was maintained with the Nosé–Hoover thermostat, while pressure regulation was achieved via the Berendsen barostat. Production simulations were conducted at 298 K and 1 atm with a 1 fs time step, extending up to 1000 ps, depending on system convergence. From the equilibrated trajectories, key transport and mechanical descriptors were extracted, including the Li^+ diffusion coefficient, ionic conductivity, density, and elastic parameters (Young's modulus, shear modulus, bulk modulus, and Pugh's ratio).

The self-diffusion coefficient (D) of the lithium ion was calculated from the mean-squared displacement (MSD) of lithium trajectories, using the following relation [54]:

$$D_i = \frac{1}{6}\frac{d(MSD_i(t))}{dt} \tag{1}$$

The mean-squared displacement (MSD) over a time interval, t is expressed as:

$$MSD_i(t) = \frac{1}{N_i}\langle \sum_{i=1}^{N_i} |r_i^c(t+t_0) - r_i^c(t)|^2 \rangle \tag{2}$$

Here, r_i^c denotes the position of the ion's center of mass at time t, and t_0 represents the time origin. The notation ⟨⋯⟩ indicates that an ensemble average is taken over multiple time origins.

The ionic conductivity (σ) was computed using the Nernst–Einstein equation [55]:

$$\sigma = \frac{e^2}{Vk_BT}\sum_{i=1}^{n} N_i z_i^2 D_i \tag{3}$$

Here, e denotes the elementary charge, V is the system volume, k_B is the Boltzmann constant, and T represents the absolute temperature. The parameter z corresponds to the valence of the lithium ion (set to 1), D refers to the self-diffusion coefficient, and N is the number of lithium ions considered in the system (assumed to be 1 in this study).

The mechanical performance of IL@PVDF systems was assessed by calculating Pugh's ratio [56]:

$$Pugh's ratio = \frac{B}{G} \tag{4}$$

where B is the bulk modulus and G is the shear modulus, derived from stress–strain analyses. A Pugh's ratio greater than 1.75 indicates ductile behavior, while values below this threshold suggest brittle characteristics.

ML modeling

To accelerate the design and optimization of high-performance IL@PVDF-based gel polymer electrolytes (GPEs) for lithium-ion batteries, a machine learning (ML) framework was developed to predict key physicochemical and mechanical properties derived from molecular dynamics (MD) simulations. The dataset comprised 103 IL@PVDF complexes, with descriptors including lithium-ion diffusion coefficient, ionic conductivity, density, Young's modulus, shear modulus, bulk modulus, and Pugh's ratio. Five supervised regression algorithms were implemented and evaluated: Random Forest (RF) [57], Gradient Boosting Regressor (GBR) [58], Extreme Gradient Boosting (XGB) [59], Light Gradient Boosting Machine (LGBM) [60], and Stacking Regressor

[61]. Tree-based ensemble methods (RF, GBR, XGB, and LGBM) were chosen for their ability to capture complex non-linear relationships, handle moderate-sized datasets robustly, and provide interpretability via feature importance analysis. The Stacking Regressor was included to combine complementary base learners, improving predictive stability and generalization across chemically diverse IL@PVDF systems. All models were trained using standardized preprocessing and hyperparameter optimization via grid search, with five-fold cross-validation performed on the training set. To prevent data leakage, scaling parameters were derived exclusively from the training set, and an external validation set (20 % of the data) was reserved for final evaluation, ensuring unbiased assessment of model generalizability. Model performance was assessed using coefficient of determination (R^2), mean absolute error (MAE), and root mean square error (RMSE).

To provide mechanistic insight into the learned structure–property relationships, SHapley Additive exPlanations (SHAP) [62–64] analysis was applied to quantify the contribution of each descriptor to the predictions. This analysis identified lithium-ion diffusion, density, and elastic moduli as the most influential features governing ionic conductivity and mechanical resilience. The integrated MD–ML framework enables high-throughput virtual screening of IL@PVDF electrolytes and facilitates the rational design of GPEs with optimized ionic transport and mechanical robustness for next-generation flexible and safe lithium-ion batteries.

Results and discussion

Distribution analysis of physicochemical and mechanical properties

To gain a foundational understanding of the variability and spread in the computed properties of IL@PVDF-based gel polymer electrolytes (GPEs), we first analyzed the distribution profiles of key descriptors derived from MD simulations (Fig. 2a–g). Figs. 2a and 2b display the distributions of lithium-ion diffusion coefficients and ionic conductivity, respectively. Both properties exhibit moderately right-skewed distributions, indicating that a subset of IL cations promote enhanced ion transport, likely due to favorable ionic radii and charge delocalization. A closer inspection of the high-performing outliers shows a tendency for cations with extended delocalized charge centers (e.g., imidazolium- and pyrrolidinium-based motifs) to facilitate ion dissociation and reduce coulombic trapping, thereby accelerating Li^+ diffusion. Fig. 2c shows the distribution of system density, which remains relatively narrow, suggesting that volumetric packing of IL@PVDF systems is not highly sensitive to cation variation. Nevertheless, slight deviations in density correlate with changes in local free volume, and systems exhibiting marginally lower densities generally align with higher diffusion coefficients, suggesting that free-volume effects partially mediate Li^+ transport. In contrast, the distributions of mechanical properties, Young's modulus (Fig. 2d), bulk modulus (Fig. 2e), and shear modulus

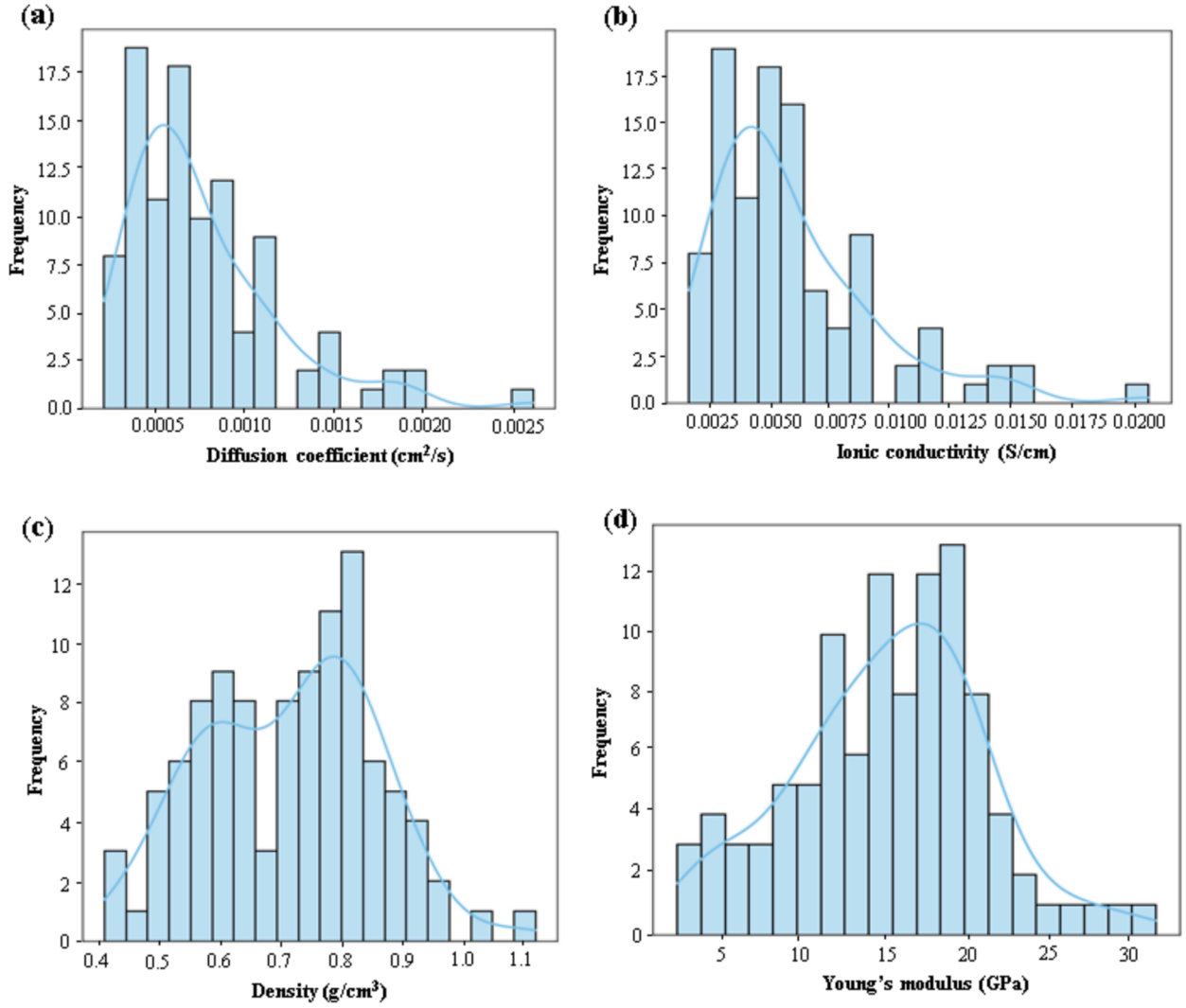


Fig. 2. Distributions of physicochemical and mechanical properties of IL@PVDF-based gel polymer electrolytes (GPEs): (a) lithium-ion diffusion coefficient, (b) ionic conductivity, (c) density, (d) young's modulus, (e) bulk modulus, (f) shear modulus, and (g) Pugh's ratio.

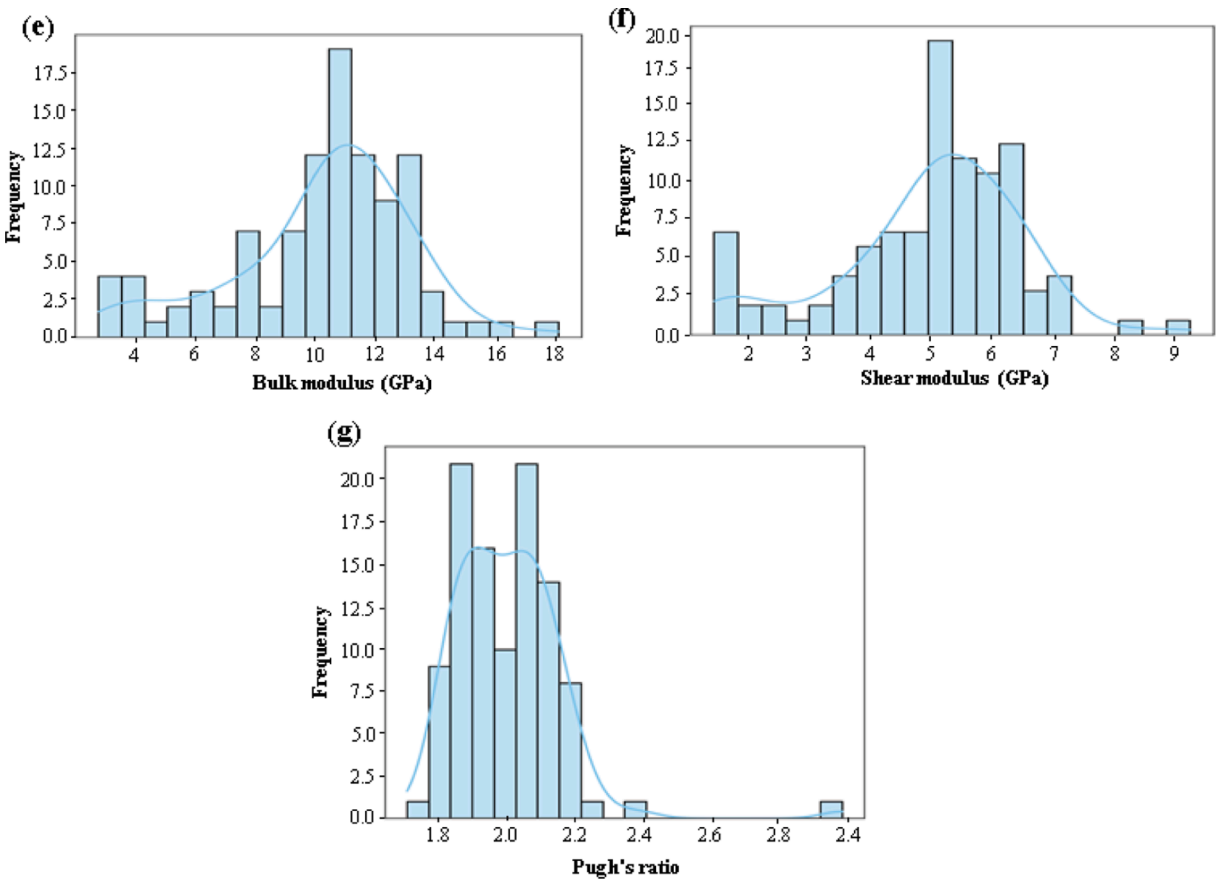


Fig. 2. (*continued*).

(Fig. 2f), are more symmetric, reflecting consistent stiffness responses across most formulations. This indicates that IL incorporation enhances ductility and reduces brittleness without severely compromising stiffness, a crucial advantage for flexible or wearable battery applications. Additionally, the distribution of Pugh's ratio (Fig. 2g) indicates that the majority of the designed IL@PVDF-based electrolytes exhibit a predominantly ductile character, with values exceeding the critical threshold of 1.75. This ductility is advantageous for flexible and mechanically robust battery components. Furthermore, the simultaneous analysis of mechanical and transport properties reveals no pronounced trade-off between them. To confirm this, correlation analyses were conducted, showing weak dependencies ($R^2 < 0.3$) between modulus values and ionic conductivity, thereby indicating that mechanical robustness and ionic mobility can be independently optimized. Similar conclusions were recently drawn by Shi et al. [65] through Onsager transport theory and MD simulations, demonstrating that transport properties in PVDF-based electrolytes can be decoupled from other physicochemical responses, offering composition-driven optimization strategies. Such decoupling suggests that cation selection can be tailored to prioritize either mechanical compliance or transport performance without incurring penalties in the other property domain, establishing a rational design principle for next-generation polymer electrolytes. Achieving this balance is critical for practical applications where both mechanical resilience and efficient ion transport are essential. Overall, the distribution analysis provides not only statistical ranges of achievable properties but also mechanistic insights and design rules, thereby laying the groundwork for predictive machine learning models to identify IL cations with synergistic electrochemical and mechanical performance.

Correlation and cation-dependent trends

Beyond examining individual property distributions, we further investigated the interrelationships among the key transport and mechanical descriptors as a function of ionic liquid (IL) chemistry. Fig. 3 presents a comprehensive correlation heatmap of the principal descriptors. Strong positive correlations were identified among the mechanical moduli (Young's, shear, and bulk), indicating that stiffness enhancement trends are consistent across different mechanical responses. In contrast, both ionic conductivity and lithium-ion diffusion coefficients exhibited weak or negative correlations with these stiffness indicators, suggesting a fundamental trade-off between mechanical robustness and ion mobility. These decouplings highlight an essential principle for electrolyte optimization, enhancing one property domain does not necessarily require compromising the other.

Fig. S2 provides a detailed comparative assessment of how cation structure, steric volume, and electronic polarizability collectively influence the ionic transport and mechanical performance of IL@PVDF-based gel polymer electrolytes (GPEs). As illustrated in Fig. S2a, imidazolium- and pyrrolidinium-based ILs exhibit significantly higher ionic conductivities compared to ammonium counterparts. This enhancement arises from their delocalized charge distributions and reduced steric hindrance, which facilitate faster ion migration. Conversely, ammonium-based systems exhibit stronger ion pairing and steric congestion, thereby limiting ionic mobility. The variation in Pugh's ratio across these cation families (Fig. S2b) further reveals that sulfonium- and pyrrolidinium-based ILs exhibit superior ductility and mechanical flexibility, emphasizing the dual impact of cation chemistry on electrochemical transport and mechanical resilience.

The box plot of ionic conductivity versus cation volume (Fig. S2c) underscores the critical role of cation size in governing charge transport.

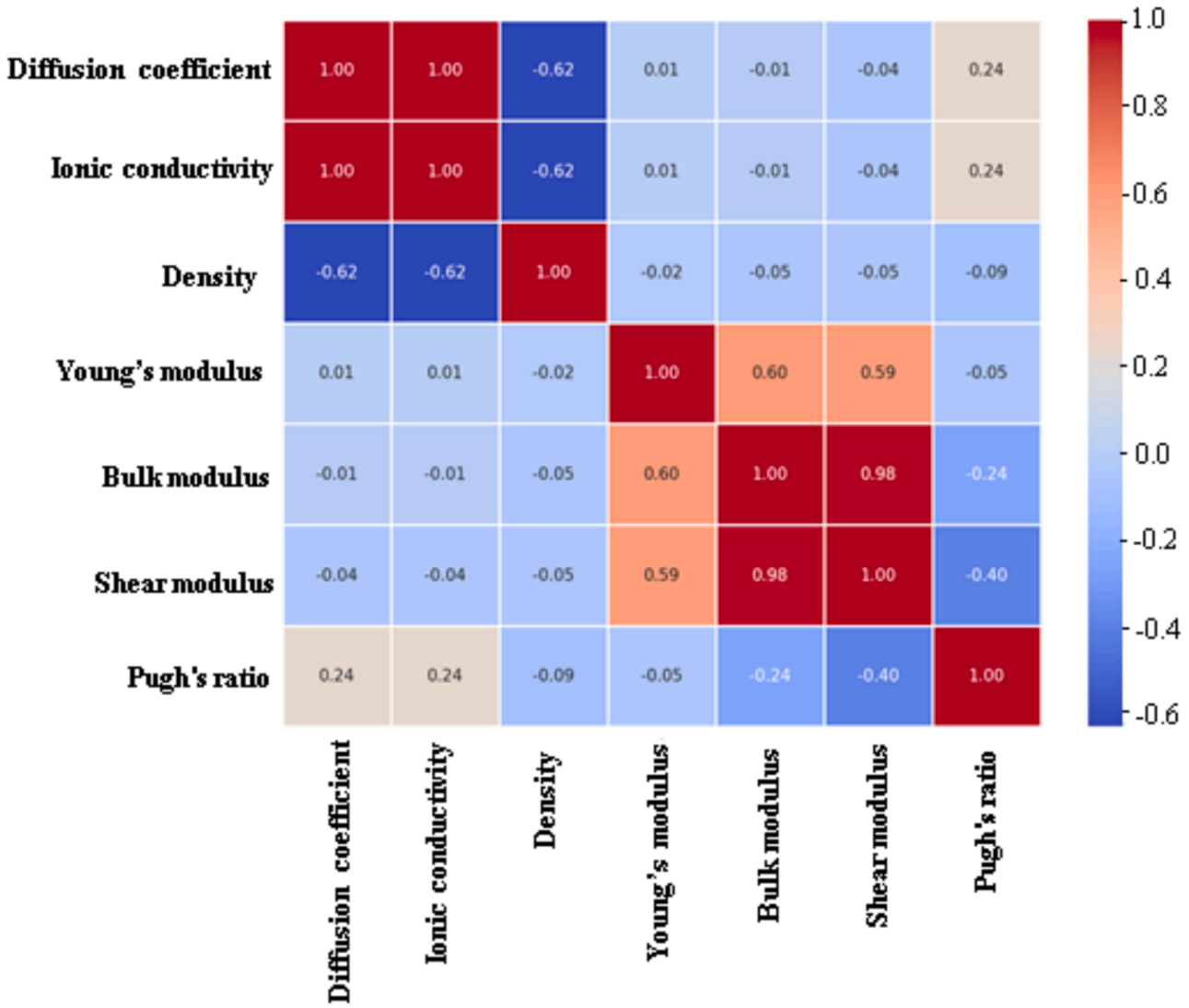


Fig. 3. Correlation heatmap between key ion transport and mechanical properties of IL@PVDF-based GPEs, revealing clear trade-offs between stiffness-related parameters (Young's, shear, and bulk moduli) and ionic conductivity.

Smaller cations generally yield higher conductivities due to their lower steric resistance and greater propensity for ion dissociation. In contrast, larger cations, characterized by extended van der Waals surfaces and higher rotational inertia, tend to form stronger ion–ion and ion–polymer associations, thereby restricting ion diffusion pathways. The observed relationship between ionic conductivity and cation polarizability (Fig. S2d) reveals a non-linear trend, wherein moderate polarizability promotes efficient ion transport by balancing electrostatic interactions and molecular flexibility, while excessively low or high polarizability reduces conductivity due to structural rigidity or localized overbinding effects.

Mechanical implications of cation steric and electronic factors are further elucidated in Figs. S2e and S2f, where cations with intermediate volumes and moderate polarizabilities demonstrate higher Pugh's ratios, indicative of an optimal ductile-to-brittle balance. In contrast, systems dominated by extreme smallness or excessive bulk, as well as very low or high polarizability, tend to exhibit mechanical brittleness or instability. Collectively, these observations establish that steric and electronic features of cations act synergistically to modulate both ionic conductivity and mechanical performance.

Machine learning model comparison for ionic conductivity prediction

To identify the most accurate and generalizable machine learning (ML) model for predicting the ionic conductivity of IL@PVDF-based gel polymer electrolytes (GPEs), five supervised regression algorithms, including Random Forest (RF), Gradient Boosting Regressor (GBR), Extreme Gradient Boosting (XGB), LightGBM (LGBM), and Stacking Regressor were evaluated. All models were trained using log-transformed conductivity values ($\log_{10}(\sigma)$) to mitigate skewness in the target distribution and enhance regression performance. Fig. 4 summarizes model performance across three key metrics: R^2 (Fig. 4a), Mean Absolute Error (MAE, Fig. 4b), and Root Mean Square Error (RMSE, Fig. 4c). Among the tested algorithms, XGB, RF, and Stacking achieved comparably high R^2 scores, with XGB slightly outperforming the others, whereas GBR exhibited moderate R^2 and LGBM performed poorly, suggesting limited capacity to capture the underlying structure–property relationships in this dataset. Analysis of error-based metrics corroborates these findings: XGB consistently yielded the lowest MAE and RMSE, demonstrating superior predictive accuracy and generalization, while GBR and Stacking exhibited intermediate errors. LGBM displayed the highest MAE and RMSE, indicating overfitting tendencies or suboptimal utilization of the combined molecular and mechanical descriptors. Collectively, these results identify XGB as the most accurate

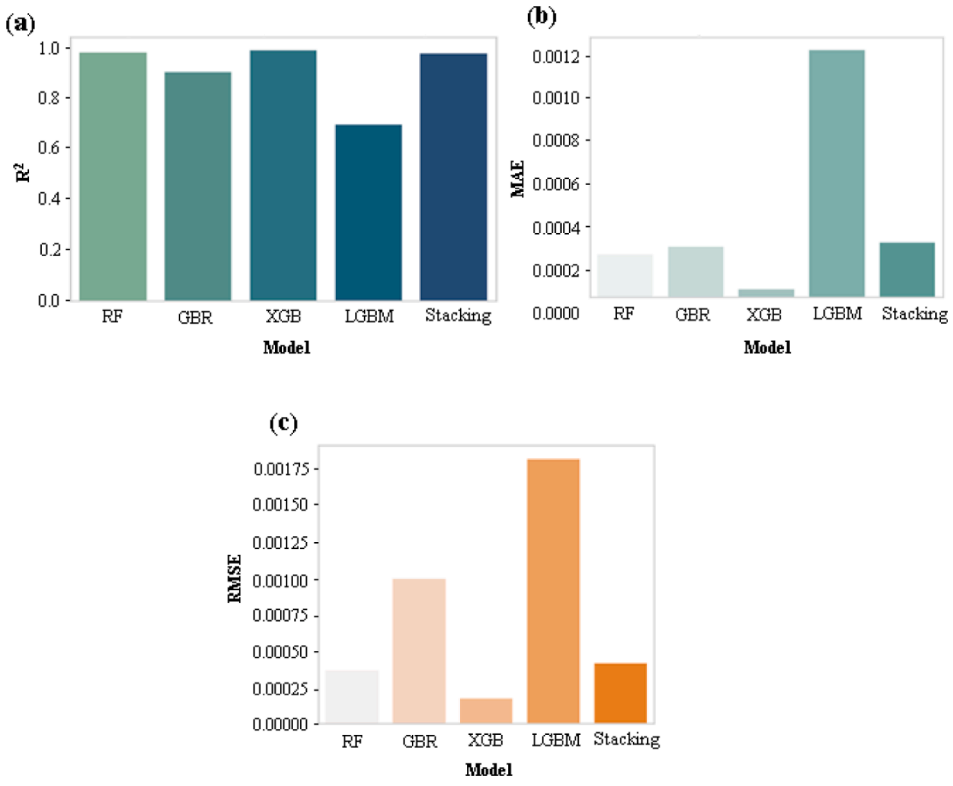


Fig. 4. Comparative performance of four supervised ML models: RF, GBR, XGB, LGBM, and stacking in predicting the log-transformed ionic conductivity of IL@PVDF-based GPEs, evaluated using (a) R^2, (b) MAE, and (c) RMSE metrics.

and reliable model for ionic conductivity prediction in IL@PVDF systems, with RF and Stacking as strong alternatives where model interpretability or ensemble stability is prioritized. Feature importance analysis further revealed that cation-related molecular descriptors, such as ionic volume, charge delocalization, and polarizability, were the primary drivers of conductivity variations, while mechanical properties contributed secondary but non-negligible effects. This finding suggests that ML models not only provide predictive capability but also offer mechanistic insights, guiding rational cation selection and IL-polymer formulation to achieve targeted transport performance. The high performance of tree-based ensemble methods highlights the presence of complex nonlinear dependencies between molecular descriptors, mechanical properties, and ion transport, underscoring the necessity of advanced algorithms for robust property prediction [66] These insights provide a systematic benchmark for future ML-driven optimization and interpretability studies, offering a pathway to accelerate screening and rational design of high-performance IL@PVDF electrolytes.

Predictive accuracy and generalization of the best model

Further evaluation of the best-performing model, XGB, is provided in Fig. 5, which illustrates its predictive accuracy, learning dynamics, and residual behavior. The actual vs. predicted plot (Fig. 5a) demonstrates a strong linear relationship between predicted and simulated ionic conductivity values, with the majority of points closely aligned along the identity line, indicating minimal bias and high predictive fidelity. This alignment underscores the model's ability to generalize across a chemically diverse range of IL@PVDF systems, effectively capturing both low- and high-conductivity regimes with high precision. The learning curve analysis (Fig. 5b) reveals that both training and validation R^2 scores improve steadily with increasing dataset size, without evidence of early plateauing or divergence between the curves, suggesting that the model is not overfitting and that the descriptor space sufficiently captures the underlying structure–property relationships. The upward trajectory of the curves further implies that predictive performance could be enhanced with additional data, highlighting the scalability of this approach to larger chemical libraries or extended MD simulations. Fig. 5c presents the residual error distribution (predicted − actual conductivity), which is symmetrically centered around zero with a narrow spread and no significant outliers, further confirming the robustness and reliability of XGB predictions across the full conductivity spectrum. Taken together, these results establish XGB as a high-performing, well-generalized regression model for predicting ionic conductivity in IL@PVDF-based GPEs, capable of capturing complex nonlinear and multivariate relationships among molecular, transport, and mechanical descriptors. Its strong performance across multiple evaluation metrics, favorable training behavior, and stable residual distribution position it as a reliable engine for high-throughput screening and rational design of advanced electrolyte materials [48,67].

To further ensure the robustness of the computational framework, the ML predictions were benchmarked against experimentally reported data for similar ionic liquid–fluoropolymer systems. For instance, Silva et al. [68] combined experimental and molecular dynamics analyses of PVDF@[EMIM][FSI] electrolytes, demonstrating agreement between simulated and measured ionic conductivities. Similarly, Chen et al. [69] reported a PVDF-HFP@[EMIM][BF_4] gel polymer electrolyte exhibiting an ionic conductivity of 6.9×10^{-3} S/cm and high thermal safety, while Serra et al. [36] developed [BMIM][SCN]/PVDF-HFP solid electrolytes with conductivities up to 0.15 mS/cm and excellent electrochemical stability. The consistency of our MD–ML predictions with these experimentally observed trends supports the validity of the employed force field and confirms that the developed model reliably captures the physicochemical mechanisms governing ion transport and mechanical response in IL@PVDF complexes.

Explainable machine learning: SHAP-based feature interpretation

To gain deeper mechanistic insights into the structure–property

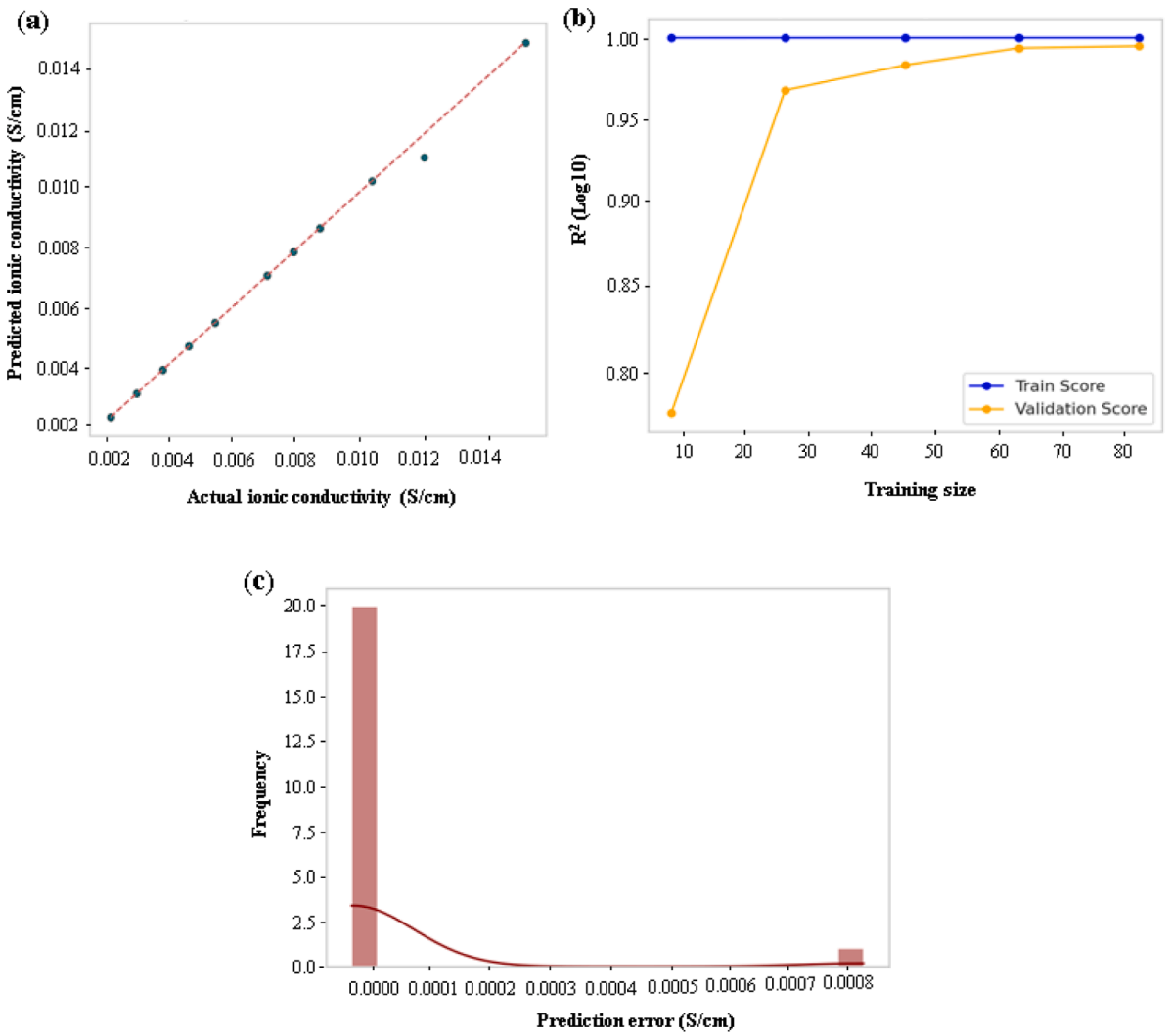


Fig. 5. (a) Actual vs. predicted ionic conductivity values for the best-performing model (XGB), (b) learning curve showing training and validation R^2 scores as a function of training set size, and (c) residual error distribution, highlighting the predictive accuracy and variance.

relationships governing ionic conductivity in IL@PVDF-based gel polymer electrolytes (GPEs), SHapley Additive exPlanations (SHAP) analysis was applied to interpret the predictions of the best-performing XGBoost (XGB) model. SHAP provides model-agnostic, locally interpretable feature attributions, enabling transparent understanding of complex ensemble model behavior [70]. Fig. S3 presents the comprehensive SHAP analysis, illustrating both the global and local effects of key physicochemical and mechanical descriptors on the predicted log-transformed ionic conductivity ($\log_{10}\sigma$). As shown in the SHAP summary (beeswarm) plot (Fig. S3a), the lithium-ion diffusion coefficient and Pugh's ratio emerge as the most influential parameters. High diffusion coefficients consistently shift conductivity predictions upward, reflecting the direct physicochemical relationship between ion mobility and ionic conductivity. Mechanical descriptors, including Pugh's ratio, shear modulus, and Young's modulus, exhibit overall negative correlations with conductivity, suggesting that stiffer polymer matrices restrict chain dynamics and thereby hinder ion migration. In contrast, moderate Pugh's ratio values, indicative of balanced ductility, facilitate microstructural flexibility, enabling dynamic rearrangements that support faster lithium-ion transport. Density also contributes moderately by influencing the free volume accessible for ionic movement.

The global SHAP feature-importance plot (Fig. S3b) reinforces these findings, showing that the diffusion coefficient dominates the conductivity prediction landscape, followed by Pugh's ratio and density. This highlights the coupled roles of dynamic ion transport and matrix elasticity. The SHAP dependence plot (Fig. S3c) further confirms a strong positive trend between conductivity and diffusion coefficient, with low variance across the data space. The SHAP interaction plot (Fig. S3d) reveals a non-linear synergistic relationship between diffusion coefficient and Pugh's ratio, demonstrating that systems with both high diffusion and moderate ductility achieve the highest predicted conductivities. Collectively, this SHAP-based interpretability analysis establishes a physically grounded understanding of the XGB model, linking ionic transport and mechanical properties directly to conductivity behavior. These insights provide actionable design principles for optimizing IL@PVDF-based GPEs, enabling the targeted enhancement of both electrochemical performance and mechanical robustness through molecular-level engineering.

Property-specific modeling

To further assess the predictive robustness and generalizability of the developed machine learning models, Fig. 6 presents a comparative evaluation of model interpretability and error distributions. Fig. 6a shows the feature importance ranking derived from SHAP analysis. The lithium-ion diffusion coefficient emerged as the most influential predictor of ionic conductivity, consistent with the physical intuition that ion mobility directly governs charge transport in GPEs. This underscores

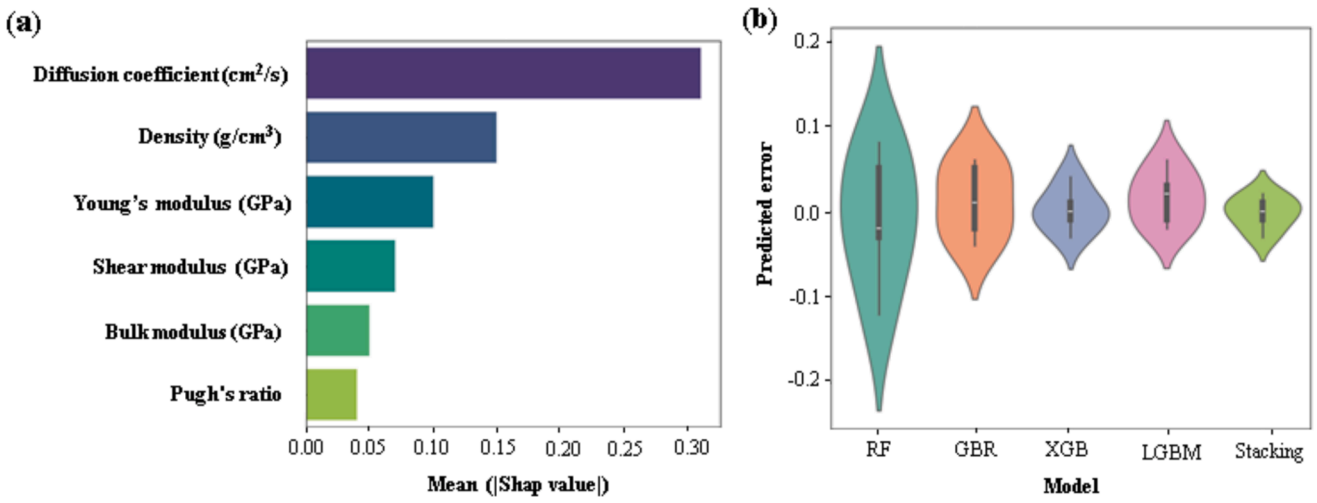


Fig. 6. (a) SHAP-based feature importance ranking for predicting ionic conductivity in IL@PVDF-based GPEs and (b) Violin plots showing the distribution of residual prediction errors for five machine learning models (RF, XGB, LGBM, GBR, and Stacking).

the critical role of microscopic ion dynamics, captured via molecular dynamics simulations, in accurate property prediction. Following the diffusion coefficient, density and Pugh's ratio ranked highly, indicating that matrix packing density and mechanical flexibility significantly influence ionic transport pathways within the IL@PVDF composite. In contrast, mechanical stiffness parameters, including bulk modulus, shear modulus, and Young's modulus, were of relatively lower importance, supporting the notion that overly rigid polymer matrices can impede ion motion and reduce electrolyte performance. Overall, the SHAP analysis provides a physically meaningful feature hierarchy, guiding future materials design by prioritizing descriptors related to ion mobility and flexible matrix architectures. Fig. 6b displays violin plots of residual prediction errors across the evaluated models. The stacking model demonstrated the narrowest residual error distribution, indicating enhanced robustness and the ability to reduce variance by combining complementary base learners. This suggests that stacking effectively smooths prediction errors and mitigates extreme deviations, even though its average performance metrics (R2, MAE, RMSE) were comparable to, but slightly lower than, those of XGB. Quantitative cross-validation results confirmed that XGB achieved the highest R2 and the lowest MAE and RMSE, affirming its strong overall predictive accuracy. The slightly tighter residual distribution of the stacking model, despite marginally lower aggregate error metrics, reflects the ensemble's strength in enhancing error consistency and reliability across diverse IL@PVDF samples, providing complementary insights to the single-model predictions.

Comparative analysis with previous MD–ML frameworks for polymer electrolytes

To contextualize the present work within the broader landscape of MD–ML studies, several representative examples of electrolyte systems and their associated modeling strategies are summarized in Table 1. Solid polymer electrolytes (SPEs) have been investigated using system behavior–based descriptors derived from molecular dynamics (MD) to predict Li-ion diffusion and ionic conductivity [71], while coarse-grained SPE models employing physically interpretable universal descriptors combined with Bayesian Optimization (BO) have been used to guide the design of highly conductive SPEs [72]. In solid-state electrolytes, Li_6PS5Cl has been studied using effective Li jump rates and local atomic environment descriptors to predict ionic conductivity [73], and $Li_7La_3Zr2O12$ (LLZO) has been modeled using neural-network descriptors trained on ab initio data to capture Li diffusivity and structural transitions [74]. For ionic liquids, molecular descriptors obtained from MD simulations have been applied to predict electrochemical windows (ECW) [75]. The present work on IL@PVDF gel polymer electrolytes (GPEs) integrates molecular descriptors, Li^+ diffusion coefficients, Pugh's ratio, and other mechanical and transport properties to predict ionic conductivity. Collectively, these studies demonstrate the versatility of MD–ML frameworks across different classes of electrolytes, highlighting the importance of descriptor selection and ML methodology in achieving accurate, physically interpretable predictions of key target properties.

Table 1
Comparative overview of machine learning–molecular dynamics frameworks applied to various electrolyte system

Electrolyte System	Features	Target Property	Reference
Solid polymer electrolytes	System behavior-based descriptors from MD	Li^+ diffusion, ionic conductivity	[71]
Solid polymer electrolytes	Physically interpretable universal descriptors derived from coarse-grained chemical species	Ionic conductivity	[72]
Li_6PS5Cl solid-state electrolyte	Effective Li jump rate, local atomic environment	Ionic conductivity	[73]
$Li_7La_3Zr_2O_{12}$ (LLZO) solid-state electrolyte	Neural-network descriptors from ab initio data	Li diffusivity, structural transition	[74]
Ionic liquid	Molecular descriptors from MD	Electrochemical window (ECW)	[75]
IL@PVDF gel polymer electrolytes	Molecular descriptors, Li^+ diffusion coefficient, Pugh's ratio, mechanical & transport descriptor	Ionic conductivity	This work

Limitations of the MD–ML framework

While the proposed MD–ML framework demonstrates strong predictive capability and mechanistic insight, several limitations should be acknowledged. The current dataset includes 103 IL@PVDF systems with a fixed alaninate anion and a limited set of cation chemistries, which may constrain generalizability to untested polymers, anions, or multi-component electrolytes. The molecular dynamics simulations rely on

the COMPASS force field and relatively short simulation times (~1 ns), which may not fully capture long-timescale diffusion events or large-scale polymer rearrangements. Additionally, while SHAP analysis provides interpretable feature contributions, some complex non-linear interactions or emergent effects may remain underexplored. Despite these limitations, the framework offers substantial benefits. By integrating MD-derived structural, transport, and mechanical descriptors with machine learning, it enables rapid, physically informed screening of IL@PVDF-based electrolytes, revealing key design principles for balancing ion transport and mechanical flexibility. The combination of predictive accuracy, interpretability via SHAP, and the ability to explore a chemically diverse set of IL cations provides actionable guidance for the rational design of next-generation gel polymer electrolytes. This approach not only accelerates the identification of high-performance electrolyte formulations but also establishes a scalable methodology for extending the framework to larger datasets, additional polymer matrices, and broader chemical spaces, offering a roadmap for future computational and experimental studies.

Conclusion

In this work, we present a synergistic framework combining molecular dynamics (MD) simulations and machine learning (ML) techniques to predict, interpret, and rationalize the ion transport and mechanical properties of IL@PVDF-based gel polymer electrolytes (GPEs) for lithium-ion batteries. Comprehensive distribution and correlation analyses revealed that lithium-ion diffusion coefficient, density, and Pugh's ratio are the primary determinants of ionic conductivity, while traditional mechanical stiffness parameters, such as Young's modulus, shear modulus, and bulk modulus, play a secondary role. Benchmarking of five supervised regression models demonstrated that Extreme Gradient Boosting (XGB) provides the highest accuracy, generalizability, and robustness for conductivity prediction, outperforming Random Forest, Gradient Boosting, LightGBM, and stacking ensembles. Explainable machine learning via SHAP analysis confirmed the physical relevance of key features identified by MD simulations and revealed complex, nonlinear synergies between ion mobility and polymer ductility, indicating that high ionic conductivity and favorable mechanical flexibility can be simultaneously achieved through careful IL cation selection and matrix design. Feature-specific analyses further highlighted the critical role of microstructural packing and polymer flexibility in modulating ion transport pathways, providing actionable design rules for next-generation electrolytes. The proposed MD–ML framework offers a scalable, high-throughput platform for screening large chemical libraries of ILs, reducing reliance on trial-and-error experimentation, and accelerating the rational design of multifunctional GPEs. By combining predictive accuracy with mechanistic interpretability, this approach bridges the gap between data-driven modeling and fundamental materials understanding, demonstrating a versatile strategy to guide electrolyte innovation and facilitate the development of mechanically robust, high-performance polymer electrolytes for advanced lithium-ion battery applications.

CRediT authorship contribution statement

Sadegh Kaviani: Writing – review & editing, Writing – original draft, Visualization, Software, Methodology, Investigation, Formal analysis, Data curation.

Declaration of competing interest

The author declares that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jiec.2025.11.034.

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